

Part 1: Determinants of 2×2 Matrices

On your own laptop, navigate to either or both of the following applet for 2×2 matrices:

- <https://www.geogebra.org/m/zqpftxx3>: With this applet, you can change the matrix entries and drag the edge vectors of the pre-image parallelogram
- <https://www.geogebra.org/m/bbfj2n99>: With this applet, you can change the matrix entries and drag the edge vectors of the image parallelogram

1. In your small group, explore the applet for 5-10 minutes. *What do you notice? What do you wonder?* Record your observations, conjectures, and questions.
2. In your small groups, create at least three different matrices that have determinant 0.
 - (a) What is the geometric impact of the various matrix transformations?
 - (b) What are some similarities and differences in these specific matrices?
 - (c) Develop some conjectures: under what conditions is the determinant zero?
3. In your small group, work together and find as many matrices as you can that have determinant 2.
 - (a) Organize your examples according to the geometric impact of the matrix transformations.
 - (b) If you have any observations or develop any ideas that can be generalized, write those on your slide, too.
4. In your small groups, create at least three different matrices that have a negative determinant.
 - (a) What is the geometric impact of the various matrix transformations?
 - (b) What are some similarities and differences in these specific matrices?
 - (c) Develop some conjectures: under what conditions is the determinant negative?

Part 2: Determinants of 3×3 Matrices

On your own laptop, navigate to either or both of the following applet for 3×3 matrices:

- <https://www.geogebra.org/m/anugjjjz>: With this applet, you can change the matrix entries and rotate the graph only
- <https://www.geogebra.org/m/eajbxvfq>: With this applet, you can change matrix entries, rotate the graph, swap rows and columns, randomize the matrix entries, and change the image parallelepiped (and thus the matrix) by dragging the edge vectors in the graph

1. Take some of your observations for 2×2 determinants and explore how they hold for 3×3 matrices.
2. Conjecture the ways the determinant of a 3×3 matrix could be 0 and why. Then explore on the applet.
3. For each of the following operations:
 - Conjecture how the determinant will be affected by the operation and why you think it will change.
 - Test your conjecture using the applet and explain what happens geometrically (take screen shots to illustrate). You may want to open a second applet window so that you can compare the “before” and “after” side-by-side
 - (a) Switch two columns.
 - (b) Switch two rows.
 - (c) Multiply a row by a constant k .
 - (d) Multiply the whole matrix by a constant k .